To find excluded value for a given rational expression in its lowest form, say $\frac{p(x)}{q(x)}$, consider the denominator $q(x) = 0$.

For example, the rational expression $\frac{5}{x-10}$ is undefined when $x = 10$. So, 10 is called an excluded value for $\frac{5}{x-10}$.

Example 3.14 Find the excluded values of the following expressions (if any).

(i) $\frac{x+10}{8x}$ (ii) $\frac{7p+2}{8p^2+13p+5}$ (iii) $\frac{x}{x^2+1}$

Solution

(i) $\frac{x+10}{8x}$

The expression $\frac{x+10}{8x}$ is undefined when $8x = 0$ or $x = 0$. Hence the excluded value is 0.

(ii) $\frac{7p+2}{8p^2+13p+5}$

The expression $\frac{7p+2}{8p^2+13p+5}$ is undefined when $8p^2+13p+5 = 0$

that is, $(8p+5)(p+1) = 0$

$p = \frac{-5}{8}$, $p = -1$. The excluded values are $\frac{-5}{8}$ and -1.

(iii) $\frac{x}{x^2+1}$

Here $x^2 \geq 0$ for all x. Therefore, $x^2+1 \geq 0+1 = 1$. Hence, $x^2+1 \neq 0$ for any x. Therefore, there can be no real excluded values for the given rational expression $\frac{x}{x^2+1}$.

Thinking Corner

1. Are x^2-1 and $\tan x = \frac{\sin x}{\cos x}$ rational expressions?
2. The number of excluded values of $\frac{x^3+x^2-10x+8}{x^4+8x^2-9}$ is _____.

Exercise 3.4

1. Reduce each of the following rational expressions to its lowest form.

(i) $\frac{x^2-1}{x^2+x}$ (ii) $\frac{x^2-11x+18}{x^2-4x+4}$ (iii) $\frac{9x^2+81x}{x^3+8x^2-9x}$ (iv) $\frac{p^2-3p-40}{2p^3-24p^2+64p}$

2. Find the excluded values, if any of the following expressions.

(i) $\frac{y}{y^2-25}$ (ii) $\frac{t}{t^2-5t+6}$ (iii) $\frac{x^2+6x+8}{x^2+x-2}$ (iv) $\frac{x^3-27}{x^3+x^2-6x}$

3 Algebra.indd 99 16-12-2020 01:05:23 PM